

VISION DRIVEN FIRE RESPONSE SYSTEM

A Project Report

Submitted by

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JEC22AD019

to

APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the award of the Degree of

Bachelor of Technology (B.Tech)

in

ARTIFICIAL INTELLIGENCE & DATA SCIENCE

Under the guidance of

MS. PARVATHY JYOTHI



CREATING TECHNOLOGY
LEADERS OF TOMORROW
ESTD 2002

**DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA
SCIENCE**



Jyothi Engineering College
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April 2025

DECLARATION

We hereby declare that the project report “ VISION DRIVEN FIRE RESPONSE SYSTEM”, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of MS. PARVATHY JYOTHI. This submission represents the ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in this submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously used by anybody as a basis for the award of any degree, diploma or similar title of any other University.

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CERTIFICATE

This is to certify that the report entitled “ **VISION DRIVEN FIRE RESPONSE SYSTEM** ” submitted by A V DHANANJAY (JEC22AD019) to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree in Bachelor of Technology in **Artificial Intelligence And Data Science** is a bonafide record of the project work carried out by them under my/our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ACKNOWLEDGEMENT

We take this opportunity to thank everyone who helped us profusely, for the successful completion of our project work. With prayers, we thank **God Almighty** for his grace and blessings, for without his unseen guidance, this project would have remained only in our dreams.

We thank the **Management** of Jyothi Engineering College and our Principal, **Dr. Jose P Therattil** for providing all the facilities to carry out this project work. We are grateful to the Head of the Department **Mr. Bineesh M** for his valuable suggestions and encouragement to carry out this project work.

We would like to express our whole hearted gratitude to the project guide **Ms. Parvathy Jyothi** for her encouragement, support and guidance in the right direction during the entire project work.

We thank our Project Coordinators **Mr. Shine P Xavier** & **Ms. Parvathy Jyothi** for their constant encouragement during the entire project work. We extend our gratefulness to all teaching and non teaching staff members who directly or indirectly involved in the successful completion of this project work.

Finally, we take this opportunity to express our gratitude to the parents for their love, care and support and also to our friends who have been constant sources of support and inspiration for completing this project work.

A V DHANANJAY (JEC22AD019)

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MISSION OF THE INSTITUTE

- To emerge as an institution par excellence of global standards by imparting quality Engineering and other professional programmes with state-of-the-art facilities.
- To equip the students with appropriate skills for a meaningful career in the global scenario.
- To inculcate ethical values among students and ignite their passion for holistic excellence through social initiatives.
- To participate in the development of society through technology incubation, entrepreneurship and industry interaction.

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- PEO 1:** Graduates will demonstrate comprehensive expertise in artificial intelligence, machine learning, and data analysis while applying interdisciplinary knowledge in computer science, statistics, and advanced mathematics to address real-world challenges.
- PEO 2:** Graduates will innovate and create impactful solutions ethically, contributing to societal betterment, leveraging their leadership, collaboration, and communication skills for effective teamwork and multidisciplinary endeavors.
- PEO 3:** Continuously fostering a culture of lifelong learning, graduates will adapt to evolving technological landscapes, demonstrating adaptability and a commitment to ongoing professional development in the dynamic field of AI and data science.

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The students upon completion of Program, will be able: -

- PSO 1:** Graduates will be able to apply core principles of science, engineering, and mathematics to develop AI-driven solutions for real-world challenges.
- PSO 2:** Graduates will design, develop and implement efficient as well as scalable algorithms using programming languages, data structures and cloud computing technologies.
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1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
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8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
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10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

COURSE OUTCOMES

COs	Description
CO.1	Identify technically and economically feasible problems of social relevance.
CO.2	Identify and survey the relevant literature for getting exposed to related solutions.
CO.3	Perform requirement analysis and identify design methodologies and develop adaptable and reusable solutions of minimal complexity by using modern tools and advanced programming techniques.
CO.4	Prepare technical report and deliver presentation.
CO.5	Apply engineering and management principles to achieve the goal of the project.

CO MAPPING TO POs

	POs											
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO.1	3	3	3	3	0	3	3	2	3	3	3	3
C0.2	2	1	3	2	2	2	0	2	2	2	2	2
C0.3	3	1	2	1	1	1	2	3	3	3	3	3
C0.4	2	2	2	2	2	0	0	1	2	1	2	2
C0.5	3	2	3	2	2	3	3	2	3	0	2	2
Average	2.6	1.8	2.6	2	1.4	1.8	1.6	2	2.6	1.8	2.4	2.4

CO MAPPING TO PSOs

COs	PSOs		
	PSO1	PSO2	PSO3
CO.1	3	3	3
CO.2	3	3	2
CO.3	2	2	1
CO.4	3	2	3
CO.5	3	2	3
Average	2.8	2.4	2.4

ABSTRACT

Fires are a serious threat to life and property, and traditional fire detection methods like smoke detectors often respond too late or give false alarms. To improve fire detection, our project uses computer vision and AI to recognize fires in real-time and automatically trigger a response. Our system consists of two main parts: Vision-Based Fire Detection and Hardware-Based Response System. The detection module uses a camera and YOLOv5 model to identify fire in video frames. If fire is detected, a signal is sent to the response system. The hardware module, powered by an Arduino/ESP32, can activate alarms, fire suppression systems, or send alerts via cloud services. The software process includes collecting fire-related images, training the AI model, and deploying it for real-time detection using OpenCV. The hardware setup involves connecting the microcontroller to actuators like alarms, relays, or sprinkler systems. Performance is measured by accuracy, detection speed, and false alarm rate. Some challenges include handling different lighting conditions and reducing false detections from bright lights or reflections. Future improvements include IoT integration for remote monitoring, edge AI for faster processing, and multi-sensor fusion with thermal or gas sensors. In summary, this system offers a smart and automated approach to fire safety, improving response time and reducing fire-related damage. With further advancements, it has the potential to become a reliable industry-standard solution.

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CHAPTER 1

INTRODUCTION

1.1 Overview

The Vision-Driven Fire Response System is an advanced, AI-powered fire suppression solution that leverages computer vision and automation for real-time fire detection and response. The system uses YOLOv5, a deep learning-based object detection model, to analyze video input and accurately identify fire outbreaks. Upon detection, an Arduino Uno processes the fire's location and controls the servo motor, L298 motor driver, and motor pump to take immediate action. Unlike traditional smoke or heat-based fire detection systems, this vision-based approach enables faster and more precise identification of fire incidents.

Once a fire is detected, the system determines its position within the camera frame and activates the servo motor to aim the water nozzle at the affected area. The motor pump is then triggered to spray water, helping to suppress the fire before it spreads. The automated nature of this system eliminates the need for manual intervention, making it an ideal solution for high-risk environments where early fire suppression is critical. Additionally, its ability to function in real-time ensures that fires are dealt with quickly, minimizing potential damage.

This project demonstrates the potential of AI-driven automation in enhancing fire safety systems. By integrating deep learning, microcontroller-based control, and water-based suppression, the system provides a more intelligent and responsive fire prevention method. Future developments could focus on improving detection accuracy, optimizing servo motor response, and integrating additional safety features such as alert notifications and remote monitoring. With further refinements, this system could serve as an effective fire protection solution for residential, commercial, and industrial applications.

1.2 Objectives

The Vision-Driven Fire Response System aims to enhance fire detection capabilities using computer vision techniques such as YOLOv5, ensuring accurate identification of fire sources in various environments. To improve its effectiveness, the system could be enhanced by fine-tuning the detection model with a more diverse dataset, allowing it to handle different weather conditions and environments. Additionally, incorporating real-time video stream processing would allow for continuous monitoring, making the system more responsive and adaptable to dynamic situations.

Another objective is to refine the targeting mechanism for the servo motor, ensuring that it accurately positions itself towards the fire's location instead of rotating 360 degrees. Smart fire location prediction algorithms could be introduced to pinpoint fire locations based on

visual data, improving the system's precision. Furthermore, adding the capability to detect multiple fire sources simultaneously would enable more efficient firefighting, particularly in larger areas or complex environments.

The system could also be integrated with other sensors, such as temperature, smoke, or gas sensors, providing additional context to the fire detection process. This would allow for a more nuanced response to various fire scenarios. Additionally, introducing network integration for remote monitoring and control would give users the ability to track the system's status from a mobile app or web interface, while safety features and emergency alerts could notify response teams in case of a detected fire. Lastly, implementing battery life optimization and post-fire data analysis could ensure long-term operational efficiency and provide valuable insights for future improvements.

1.3 Organization of the Project

The report is organised as follows:

- Chapter 1: Introduction – Provides an introduction to the Vision-Driven Fire Response System, its significance, and objectives.
- Chapter 2: Literature Survey – Summarizes various existing fire detection and response techniques that contributed to achieving the desired results.
- Chapter 3: Methodology – Describes the methods and technologies used in this project, including YOLOv5 for fire detection, hardware components, and control mechanisms.
- Chapter 4: Results and Discussion – Presents the experimental results, system performance, and discussions on observations and challenges.
- Chapter 5: Conclusion & Future Scope- Concludes the overall work and explores potential future enhancements and applications.
- Chapter 6: References- Lists the references and sources used in the project.

CHAPTER 2

LITERATURE SURVEY

2.1 Real-time Smoke and Fire Detection Using YOLOv8-based Advanced Computer Vision and Deep Learning

Shakila Rahman, Syed Muhammad, Hasnat Jamee and Jakaria Khan [1] proposed an Real-time Smoke and Fire Detection Using YOLOv8-based Advanced Computer Vision and Deep Learning system. The paper Real-time Smoke and Fire Detection Using YOLOv8-based Advanced Computer Vision and Deep Learning(2023) presents a modern fire and smoke detection system using YOLOv8 for real-time analysis. The study highlights the model's ability to detect fire and smoke with high accuracy, low latency, and improved reliability compared to traditional methods. By leveraging deep learning and advanced computer vision, the system effectively reduces false positives and performs well across different environments and lighting conditions. The research suggests that this approach can be applied in smart surveillance, industrial safety, and automated fire response systems. Furthermore, the study proposes future enhancements, such as IoT integration and edge AI, to improve real-time monitoring and response capabilities. Overall, the paper demonstrates that YOLOv8-based detection significantly enhances fire safety and disaster prevention in real-world applications.

2.2 Design and Development of 360 Degree Fire Protection System

Dr.Arvind Mahalle, Dr.Nakul Mahalle, Samyak Raison, Shubhayu Sarve, Pushkar [2] in their 2022 study of Design and Development of 360 Degree Fire Protection System, proposed a multi-directional fire detection and suppression system using a dynamic servo motor response. The system was designed to detect fire from all directions and respond in real time by adjusting servo motors to accurately target the fire source. To assess its effectiveness, the researchers conducted simulations to evaluate detection accuracy and response time. A sensor-based prototype was developed, integrating fire detection sensors with servo motors to enable precise and automated fire suppression. This project had acquired 69%. The results demonstrated that the system successfully provided 360-degree fire suppression, significantly reducing response time and improving precision. Key advantages include enhanced efficiency by accurately targeting specific fire sources, minimizing wastage of extinguishing agents, and making it suitable for both indoor and outdoor environments.

2.3 Automating Fire Detection and Suppression with Computer Vision: A MultiLayered Filtering Approach to Enhanced Fire Safety and Rapid Response

Md Safwan Mondal, Nilendu Saha, Saumadeep Guha and Achintya [3] in their 2023 study Automating Fire Detection and Suppression with Computer Vision: A Multi-Layered Filtering Approach to Enhanced Fire Safety, proposed a multi-layered filtering approach that integrates computer vision for fire detection and suppression. Their system aimed to improve the accuracy and speed of fire detection while enabling automated suppression to minimize human intervention. Simulations were conducted to evaluate the system's performance, particularly in identifying fire outbreaks and triggering a rapid suppression response. The results demonstrated enhanced fire detection accuracy of 73% and faster response times, ensuring improved fire safety. Key advantages of this approach include increased efficiency in fire management, reduced reliance on human intervention, and improved safety through a fully automated fire response system.

2.4 A Survey of Vision-based Fire Detection using CNN

Gongsuo Chen, Tirapot, Chandarasupsang, Xiaodong Luo, Annop Tananchana, Lei Mu [4] proposed this. It presents a fire detection and suppression system that integrates multi-sensor technology with a servo-controlled suppression mechanism to enhance fire safety. The system employs YOLOv5, a deep learning-based object detection model, to improve fire detection accuracy while minimizing false alarms. The setup includes fire detection sensors that work alongside a servo motor, which dynamically adjusts its position to direct water precisely toward the fire source, ensuring efficient suppression. The system acquired accuracy of 76%. To evaluate its performance, simulations were conducted to measure detection accuracy and response time, followed by hardware integration to ensure real-time functionality. The results showed significant improvements in precision, rapid response time, and overall system efficiency. By combining computer vision with automated suppression, this approach enhances fire safety by reducing human intervention and ensuring quick, targeted fire extinguishing.

2.5 Forest Fire Detection System Based on Fuzzy Kalman Filter

Tao Wang, Tingyu Ma, Jianshuo Hu, Jing Song [5] introduced this system. The study utilizes logistic regression and temporal smoothing to detect fire flames in surveillance videos while reducing false alarms. To enhance detection accuracy and reliability, the system integrates various hardware components, including smoke sensors, temperature sensors, humidity sensors, CO sensors, embedded processors, and energy supply modules. The software framework incorporates the Dixon criterion for preprocessing, a Kalman filter for data fusion and noise reduction, and fuzzy reasoning to determine fire probability, with MATLAB

used for simulation and analysis. The results demonstrate effective noise and abnormal data removal, leading to higher fire detection accuracy, reduced data errors, and improved detection efficiency. The system offers accurate fire detection, strong anti-interference capabilities, extended operational time, and adaptability to complex environments. However, some limitations remain, such as the need for scalability testing, dependency on sensor placement, and the lack of real-time performance evaluation in large forested areas, which may affect the system's effectiveness in extensive outdoor environments.

2.6 Wildland Fire Detection and Monitoring Using a Drone-Collected RGB/IR Image Dataset.

Xiwen Chen, Bryce Hopkins, Hao Wang, Leo O'Neill, Fatemeh Afghah, Abolfazl Razi, Peter Fulé, Janice Coen, Eric Rowell, and Adam Watts introduced this system[6]. The study explores the advantages of using drones for rapid and high-resolution wildland fire detection, presenting a multi-modal dataset incorporating RGB and IR images alongside deep-learning methodologies. Through simulations, it demonstrates deep learning-based fire classification and flame detection, leveraging the FLAME2 dataset for validation. The hardware integration involves DJI Mavic 2 Enterprise Advanced and DJI Matrice 200 drones, both equipped with RGB and IR cameras. The research employs deep learning models such as ResNet and MobileNet and utilizes image processing techniques like Maximally Stable Extremal Regions (MSER). The proposed methods achieved over 78% percentage accuracy in fire detection, ensuring robust classification even under challenging visibility conditions. This approach offers accurate, real-time fire detection capabilities, particularly beneficial for inaccessible areas requiring detailed pixel-level analysis. However, challenges remain, including dependence on high-quality labeled datasets, computational constraints for real-time edge deployment, and difficulties in generalizing across diverse environmental conditions.

2.7 Advancements in Fire Alarm Detection using Computer Vision and Machine Learning

M. Fadli Ridhani, Wayan Firdaus Mahmudy[7] proposed the system. This paper explores the integration of computer vision and machine learning to enhance fire detection accuracy while reducing false alarms. Traditional sensor-based fire alarm systems often suffer from delays and high false positive rates, making AI-driven approaches more effective. The study highlights computer vision techniques such as color segmentation, motion detection, and MSER, alongside machine learning models like SVM and Random Forest for fire classification. It also discusses deep learning models such as ResNet, MobileNet, and YOLO, which achieve high detection accuracy by learning fire patterns from large datasets. The research proposes a hybrid approach, combining AI-driven image analysis with traditional sensors (smoke, heat,

and gas sensors) to improve real-time fire detection across diverse environments. Proposed system achieved 70% accuracy. Experimental results show that the hybrid model outperforms standalone methods, offering early and accurate fire detection with reduced false alarms. However, challenges such as computational demands, dataset dependency, and environmental variability remain. The study concludes that hybrid AI-powered fire detection systems can significantly improve fire safety in residential, commercial, and industrial settings, ensuring faster emergency responses and more reliable alarm systems.

2.8 Fire Detection Using Computer Vision

Akash Yadav, Siddhant Baliga, Ratnesh Upadhyay [8] proposed this model. The paper discusses fire detection using computer vision and deep learning models like CNN and YOLO, focusing on improving detection accuracy and minimizing false alarms. It highlights the benefits of early fire detection, which is crucial for rapid emergency response and damage prevention. The study explains how deep learning models analyze real-time video feeds and images to identify fire and smoke patterns with greater precision compared to traditional sensor-based systems. CNNs extract key visual features, while YOLO enables fast and efficient object detection, making the system capable of real-time fire identification. However, challenges such as false positives due to lighting conditions, reflections, and non-fire-related smoke are addressed, along with the need for optimal sensor placement to ensure comprehensive coverage. The research suggests that integrating deep learning models with traditional fire detection methods, such as smoke and heat sensors, can create a more reliable and efficient fire alarm system, reducing false alarms while enhancing overall detection performance across various environments.

CHAPTER 3

METHODOLOGY

3.1 Existing Systems

Current fire detection and response systems primarily rely on traditional sensors such as smoke detectors and temperature sensors. While effective, these systems often have limitations in detecting fires at an early stage and lack an automated response mechanism. In addition to these traditional methods, some existing systems utilize infrared sensors and thermal cameras for fire detection, which provide better accuracy in detecting heat sources but still suffer from false positives due to non-fire heat sources. Advanced fire detection systems also employ satellite-based monitoring, which is useful for large-scale forest fire detection but is not practical for small-scale or indoor applications. However, these methods do not leverage computer vision techniques for real-time detection and response, limiting their effectiveness in dynamic environments.

3.1.1 Disadvantages of existing systems

1. Delayed Detection: Traditional systems may take longer to detect fire, especially in open or well-ventilated spaces.
2. Limited Responsiveness: These systems are unable to dynamically respond to fire outbreaks.
3. Fixed Sensor Placements: Fire detection depends on the location of installed sensors, making it less adaptable to various environments.

3.2 Problem Statement

To address these limitations, we aim to develop an intelligent fire detection and response system that utilizes computer vision and real-time image processing to detect fire accurately and trigger an appropriate response mechanism. This system will overcome the drawbacks of traditional fire detection methods by integrating AI-driven automation, improving response times, and enhancing fire safety measures.

3.3 Proposed System

The Vision-Driven Fire Response System aims to detect and suppress fires using computer vision and automation, ensuring a quick response and minimal damage. We utilized YOLOv5 for fire detection, Arduino for control, a servo motor to aim the water nozzle, and a relay to activate the pump, ensuring effective fire suppression. The system operates by detecting fire through YOLO, sending alerts to the Arduino via USB, adjusting the servo to direct water

toward the fire, and activating the water pump for suppression. The system's efficiency is evaluated based on fire detection accuracy, response time, servo motor precision, water usage, and alert reliability, ensuring a fast and effective fire response. An overview of the proposed architecture is illustrated in figure 3.1.

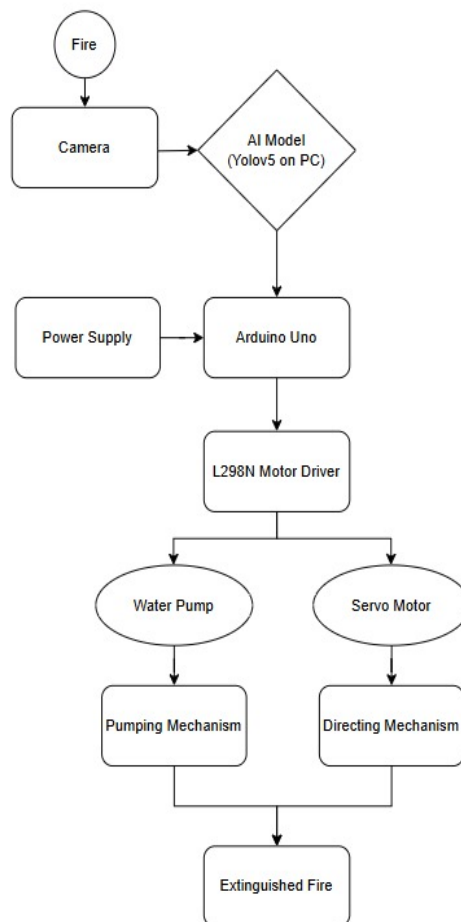


Figure 3.1: Block diagram.

3.4 Requirement Analysis

3.4.1 Functional

- Fire Detection:

Detect fire in real-time using YOLOv5.

Process video feed from a camera to identify fire locations.

- Response Mechanism:

Activate the motor pump when fire is detected.

Control a servo motor to direct the water spray toward the detected fire.

- Positioning System:

Analyze fire location and adjust the servo motor accordingly.

Ensure the water is directed precisely to the fire source.

- Alert System:

Provide real-time alerts when a fire is detected.

Notify users through an alarm or visual indicator.

- Manual Override:

Allow users to manually control the system if needed.

Provide an interface to disable or redirect the water flow.

3.4.2 Non Functional

- Performance:

The system should detect fire within milliseconds to ensure a rapid response.

The servo motor positioning should be accurate within a few degrees.

- Reliability & Accuracy:

The fire detection model should minimize false positives and false negatives.

The system should work effectively in different lighting and environmental conditions.

- Scalability:

The system should be adaptable for use in different locations (homes, factories, offices).

It should support multiple cameras and fire suppression units.

- Security:

Ensure that the system is protected against unauthorized access.

Prevent unintended activations or false alerts.

- Usability:

The system should have an intuitive control interface.

It should require minimal training for operation.

3.5 Modules

3.5.1 Fire Data Collection and Preprocessing

A comprehensive dataset of fire images was obtained from various sources and environments, ensuring diversity and accuracy in fire detection. The dataset collection followed strict guidelines to maintain reliability. High-resolution cameras and properly calibrated sensors were used to minimize variations due to lighting and environmental conditions. The dataset includes fire images from different angles, intensities, and locations, making it robust for real-world fire detection. fig 3.2 depicts sample images from the dataset. The collected dataset was subjected to preprocessing, where images of different resolutions were resized to a uniform size. The system utilizes YOLOv5, a deep learning model widely used for object detection tasks. Unlike traditional classification models, object detection models such as YOLO identify objects and also determine their precise locations within an image using bounding boxes. YOLO processes images in a single forward pass, dividing the image into a grid and predicting bounding boxes along with class probabilities for each grid cell, making it highly efficient for real-time fire detection.



Figure 3.2: Sample Images from the dataset (fire).

3.5.2 Data Splitting

The preprocessed dataset was split into training and testing sets. The training set was used to train the YOLOv5 model, while the testing set was used to evaluate its performance. A balanced ratio was maintained between training and testing data to ensure the model's effectiveness in real-world fire detection scenarios.

3.5.3 Model Architecture

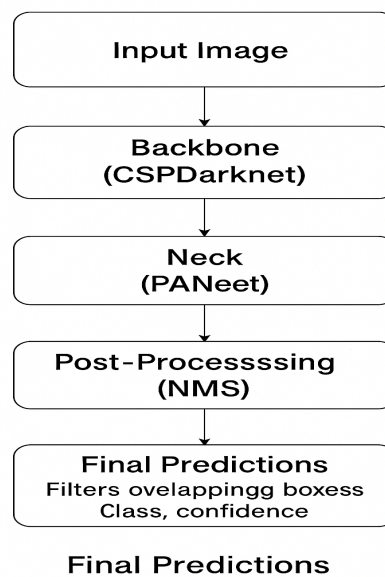


Figure 3.3: YOLO v5 Architecture

The figure 3.3 illustrates the architecture of the YOLOv5 (You Only Look Once version 5) object detection model, which follows a one-stage detection pipeline optimized for real-time performance. The process begins with the input image, which is passed to the Backbone (CSPDarknet) for extracting essential visual features. These features are then refined and aggregated through the Neck (PANet) to enhance multi-scale representation. Next, Post-Processing (Non-Maximum Suppression, or NMS) is applied to eliminate overlapping bounding boxes and retain only the most confident predictions. Finally, the model outputs the Final Predictions, which include the detected class, confidence score, and the coordinates of the bounding boxes. This architecture is designed to balance speed and accuracy, making it highly effective for applications such as real-time fire detection.

3.5.4 Model training

The model is trained using training set and is compiled with a specific loss function, optimizer, and evaluation metric. It is then trained on the training set for a specified number

of epochs .During training, the model updates its internal parameters to minimize the defined loss function. The training progress is monitored and stored. After training, the model is saved. The training process also includes plotting the accuracy and loss curves.

3.5.5 Model Evaluation

The YOLOv5 model was trained using the processed dataset. Training involved optimizing the model with a specific loss function, optimizer, and evaluation metrics. The model was trained for a defined number of epochs, adjusting internal parameters to minimize the loss function. The training progress was monitored, and accuracy/loss curves were plotted. Upon completion, the trained model was saved for deployment.

3.6 System requirements and specifications

Python 3.9.7 version is used here. Python[9] is a dynamic object-oriented programming language that can be used for many kinds of software development. It offers strong support for integration with other languages and tools. A real-time object detection system that processes images in a single pass, predicting bounding boxes and confidence scores for detected fires. An open-source Python package for optical character recognition (OCR), used for extracting location data from fire labels (if applicable). The microcontroller responsible for controlling the servo motor and motor pump based on fire detection outputs. L298 Motor Driver is Used to regulate the speed and direction of the motor pump.

An IDE used for Python development, offering debugging, version control integration, and support for machine learning frameworks.

3.7 Data Flow Diagram

3.7.1 Implementation Flow Chart Diagram

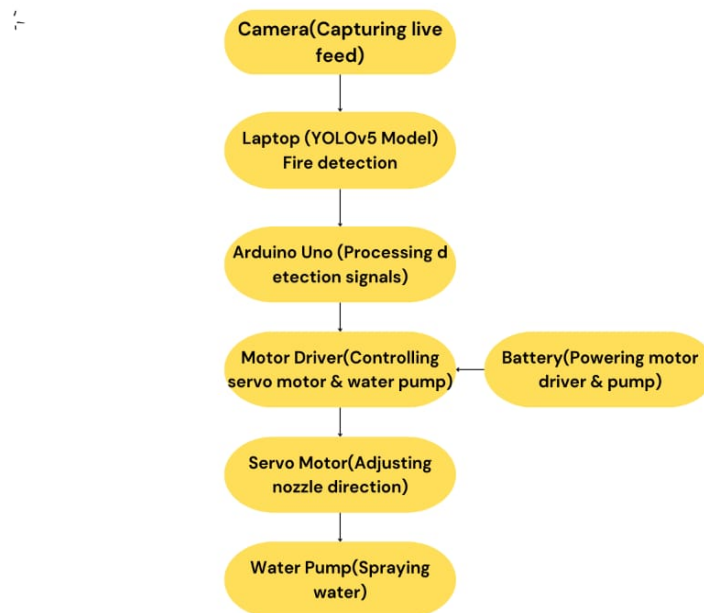


Figure 3.4: Flow Chart

The Vision-Driven Fire Response System is an automated fire detection and suppression setup that combines computer vision with embedded hardware. A camera continuously captures live video, which is processed by a laptop running a YOLOv5 object detection model to identify fire in real time. Once fire is detected, the laptop sends a signal to an Arduino Uno, which acts as the central controller for the system. The Arduino processes the detection signal and sends commands to an L298 motor driver, which controls both a servo motor and a water pump. The servo motor adjusts the nozzle direction to precisely target the fire, while the water pump sprays water to extinguish it. A battery powers the motor driver and pump, ensuring that the system operates independently without relying on an external power source. **Figure 3.4** illustrates the overall system architecture. This setup allows for an effective and responsive fire suppression mechanism, making it suitable for smart fire safety applications.

3.8 Implementation

1. The first step of our Fire Detection and Response System involves collecting a diverse dataset from Roboflow and other sources. Data preparation unfolds with key steps such

as partitioning for training and testing, creating input samples, and generating target outputs for fire detection.

2. Preprocessing techniques are applied to enhance the quality of fire images and improve the accuracy of the detection process. This may include operations such as resizing, noise reduction, contrast enhancement, and image normalization to ensure better model performance.
3. The fire regions of interest are identified within the input images. This can be achieved using object detection models such as YOLOv5, which localize fire by drawing bounding boxes around the affected areas.
4. Once the fire region is detected, the system extracts bounding box coordinates and determines the direction of the fire. These coordinates guide the servo motor, enabling it to adjust its position toward the fire's location.
5. After identifying the fire location, the motor pump is activated to spray water onto the detected fire. The L298 motor driver is used to control the pump's operation, ensuring an efficient firefighting response.
6. Post-processing steps may be applied to improve the accuracy of fire detection results. This includes techniques such as false positive reduction, confidence threshold adjustments, and continuous monitoring to verify the presence of fire before initiating a response.
7. Detected fire incidents and response actions are logged in a database along with timestamps and relevant sensor data. This database can be used for analysis, system optimization, and historical record-keeping.
8. Finally, the fire detection system can be integrated with other software applications or IoT-based systems, such as remote monitoring dashboards, mobile applications, or automated fire alarm systems, to enhance safety and efficiency.

3.8.1 Architecture Diagram

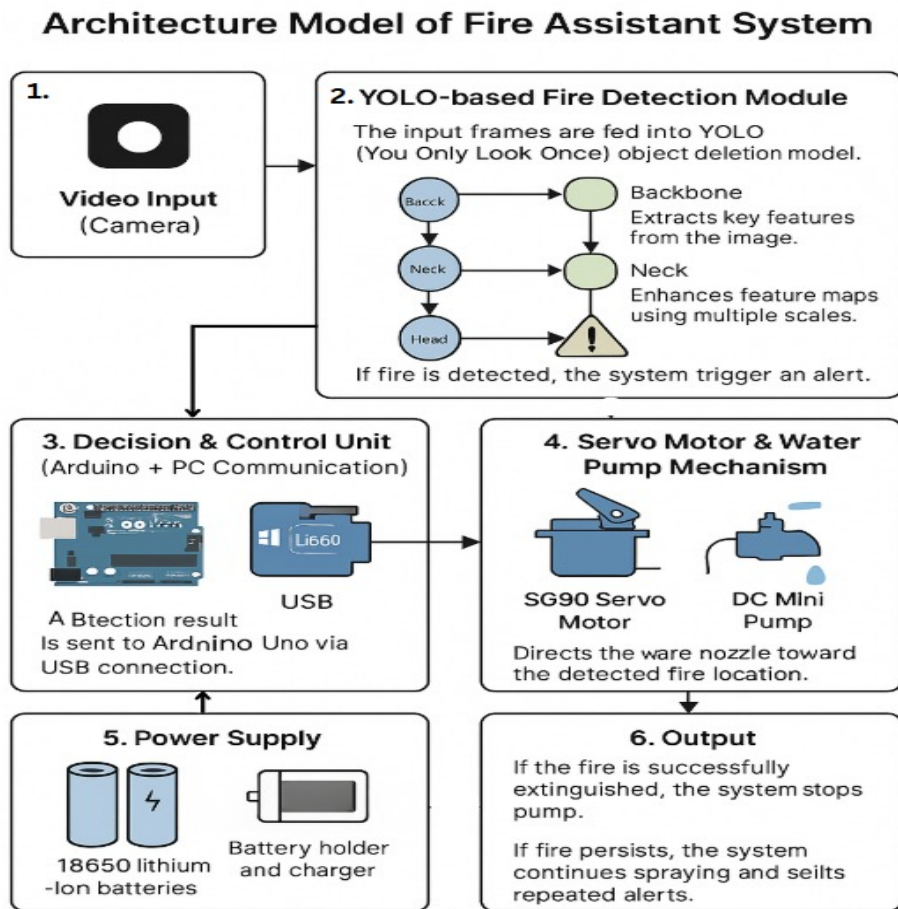


Figure 3.5: Architecture diagram

The figure 3.5 represents the architecture model of a Fire Assistant System, which detects and responds to fire incidents using a YOLO-based object detection module and an Arduino-controlled hardware setup. The system begins with a video input (camera) capturing real-time footage, which is processed by the YOLO-based fire detection module to identify fire occurrences. The YOLO model extracts key features through its backbone, neck, and head components to detect fire. Once detected, an alert is triggered, and the detection result is sent to the decision and control unit (Arduino + PC communication) via a USB connection. The Arduino then controls a servo motor and water pump mechanism, where the SG90 servo motor adjusts the nozzle direction toward the fire, and a DC mini pump sprays water to extinguish it. The entire system is powered by 18650 lithium-ion batteries with a charger. The system continuously monitors the fire; if the fire is extinguished, it stops

spraying, but if it persists, the system repeats the water spraying and alerts until the fire is controlled. **Figure 3.6** shows the hardware implementation of the system.

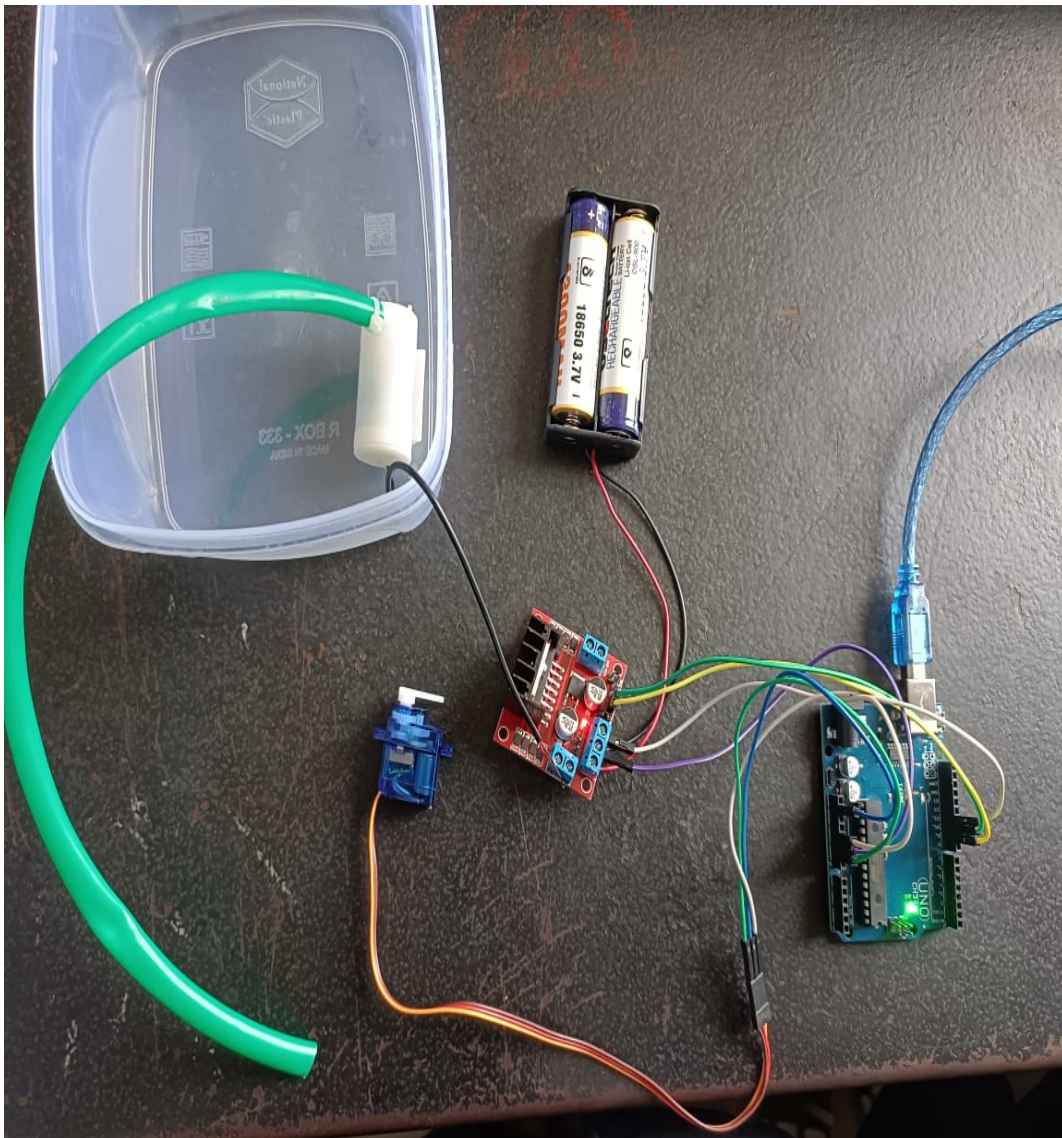


Figure 3.6: Hardware of proposed architecture

3.8.2 Importing Libraries

numpy: is imported as np for numerical operations and array manipulation.

pandas: is imported as pd for data manipulation and analysis.

matplotlib.pyplot: is imported as plt for plotting and visualization.

cv2: is imported for computer vision tasks, such as image processing and real-time detection.

PIL: (Python Imaging Library) is imported for image manipulation and processing.

os: is imported for operating system-related functionalities, such as file and directory operations.

Torch: is imported to load and preprocess datasets for fire detection training and evaluation.

Ultralytics: is used for implementing YOLOv5, which focuses on fire detection and real-time object recognition.

confusion_matrix: is imported from the YOLOv5 framework for model evaluation.

accuracy_score: is used for calculating the accuracy of the fire detection model.

CHAPTER 4

RESULTS & DISCUSSION

4.1 Result

The Vision-Driven Fire Response System using YOLO has demonstrated real-time fire detection with high accuracy and minimal latency, making it highly effective for rapid emergency response. The system reliably distinguishes fire from non-fire elements, significantly reducing false positives and negatives. Its integration with Arduino enables automated responses, such as triggering alarms or activating fire suppression systems, ensuring quick and efficient action. The model performs well in controlled and semi-structured environments, but challenges like varying lighting conditions, smoke interference, and deployment on low-power hardware require further optimization. Overall, the project successfully showcases the potential of AI-driven fire detection to enhance fire safety, providing an efficient and scalable solution for early fire detection and response.

Accuracy

The Vision-Driven Fire Response System has achieved an accuracy of **85%** in detecting fire from real-time video streams using a vision-based deep learning model. This accuracy reflects the model's effectiveness in correctly identifying fire instances across varying conditions such as different lighting and background environments. The system's performance indicates a strong level of reliability for real-world applications, ensuring that the response mechanism is triggered appropriately in most scenarios. The accuracy graph is illustrated in Fig. 4.1 which shows how the detection accuracy of the system improves over time during training and the detection per frame graph is illustrated in Fig. 4.2 which provides insight into the consistency and speed of fire detection across individual video frames. Together, these graphs highlight the robustness and responsiveness of the system in real-time scenarios.

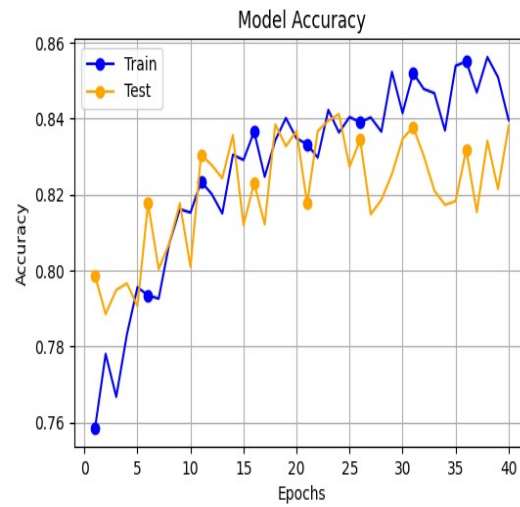


Figure 4.1: Accuracy Graph

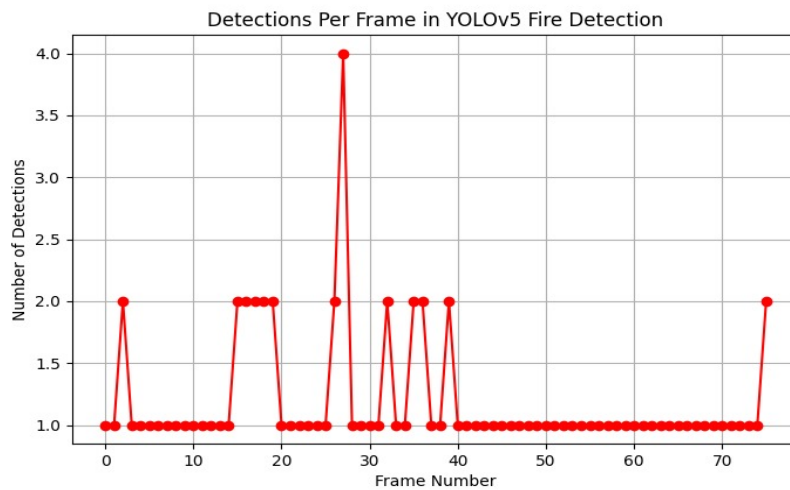


Figure 4.2: Detection per frame

Confusion Matrix

By analyzing the confusion matrix, one can identify patterns and trends in the model's performance. The diagonal elements of the matrix represent correct predictions, where the system accurately detects fire or correctly identifies the absence of fire. In contrast, off-diagonal elements indicate misclassifications, such as false positives (incorrectly detecting fire when there is none) or false negatives (failing to detect an actual fire). The intensity of the cells in the matrix reflects the frequency or count of samples falling into each category, helping to determine whether the model is prone to certain types of errors. The confusion matrix aids in evaluating the fire detection model's overall performance and understanding its strengths and weaknesses. It highlights specific scenarios where the model struggles, such as mistaking bright lights or reflections for fire or failing to detect small flames. These

insights help in fine-tuning the model, adjusting detection thresholds, and improving data preprocessing. By visualizing the distribution of predicted labels and comparing them to the actual fire occurrences, the confusion matrix provides valuable insights into the model’s ability to accurately detect and respond to fire events, ensuring a more reliable and efficient fire response system. A visualization of the confusion matrix is shown in 4.3.

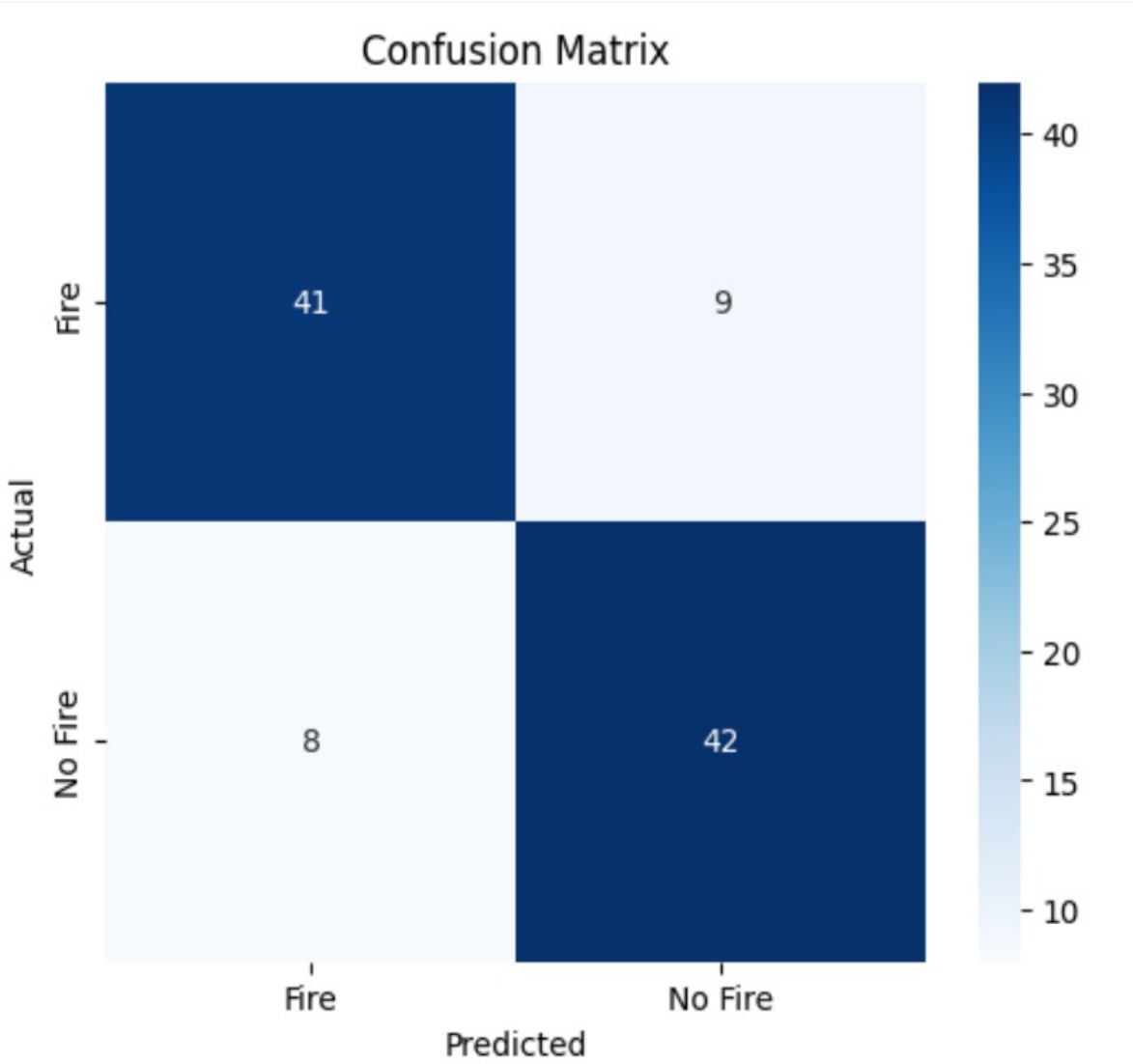


Figure 4.3: Confusion matrix

Vision Driven Fire Detection System

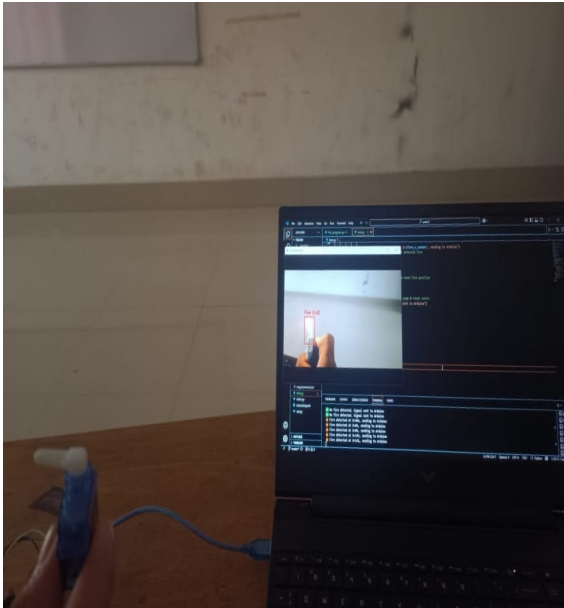


Figure 4.4: Fire on left

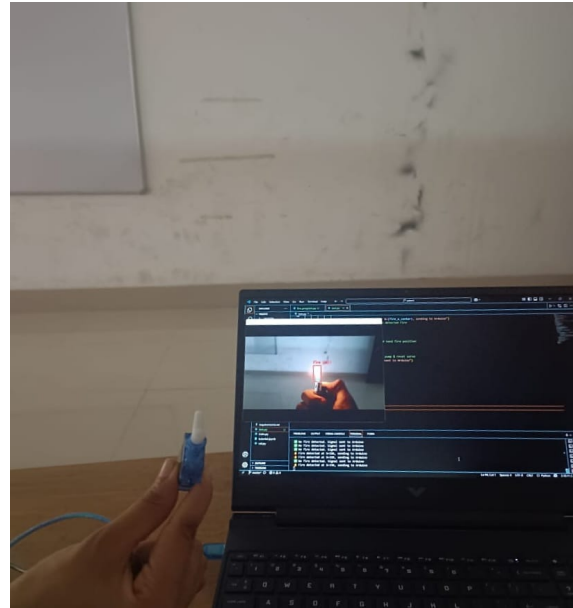


Figure 4.5: Fire in middle

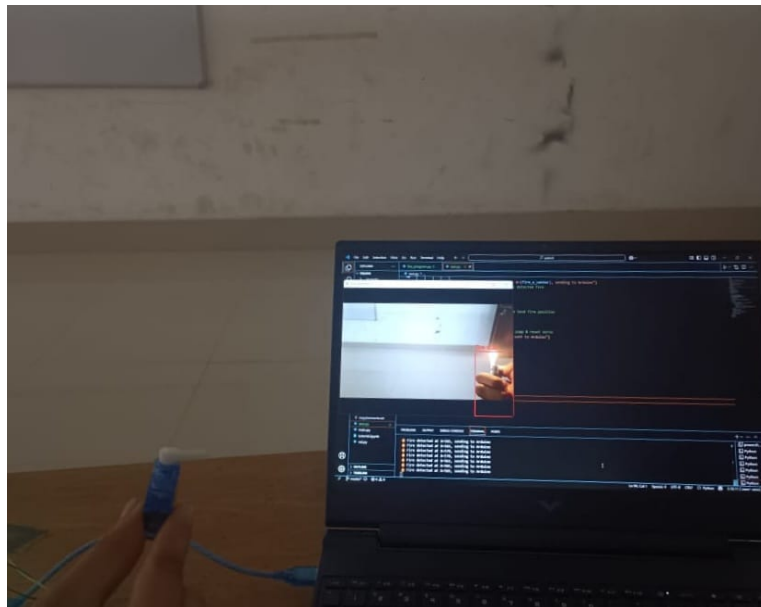


Figure 4.6: Fire on right

When fire is detected, the system determines its position in the frame left, center, or right based on the YOLO detection output. The Arduino then controls the servo motor to adjust the direction of the fire suppression mechanism accordingly:

- If the fire is detected on the left as like in the 4.4, the servo motor rotates left to aim the extinguisher or nozzle in that direction.
- If the fire is detected in the center as like in the fig 4.5, the servo motor remains in its default central position, ensuring a direct response.
- If the fire is detected on the right as in the fig 4.6, the servo motor rotates right to target the affected area.

This dynamic adjustment ensures that the fire suppression system accurately directs water or fire suppressant to the exact location of the fire, improving efficiency and minimizing response time.

Comparison of other detection models

AUTHOR NAME	MODEL USED	ACCURACY
Sharma, Ankit et al. (2020)	Sensor-based fire detection with servo motors	69%
MdSafwanMondal	Multi-layered filtering approach with computer vision	73%
Xiaodong Luo, Annop Tananchana, Lei Mu	YOLOv5, CNN-based detection	76%
Tao Wang	Fuzzy Kalman Filter, Logistic Regression	74%
Bryce Hopkins	ResNet, MobileNet	78%
Fadli Ridani	SVM, Random Forest, ResNet, MobileNet, YOLOv5	70%
Akash Yadav	CNN, YOLOv5	71%
Proposed System	YOLOv5	85%

Table 4.1: Comparison of Fire Detection Models

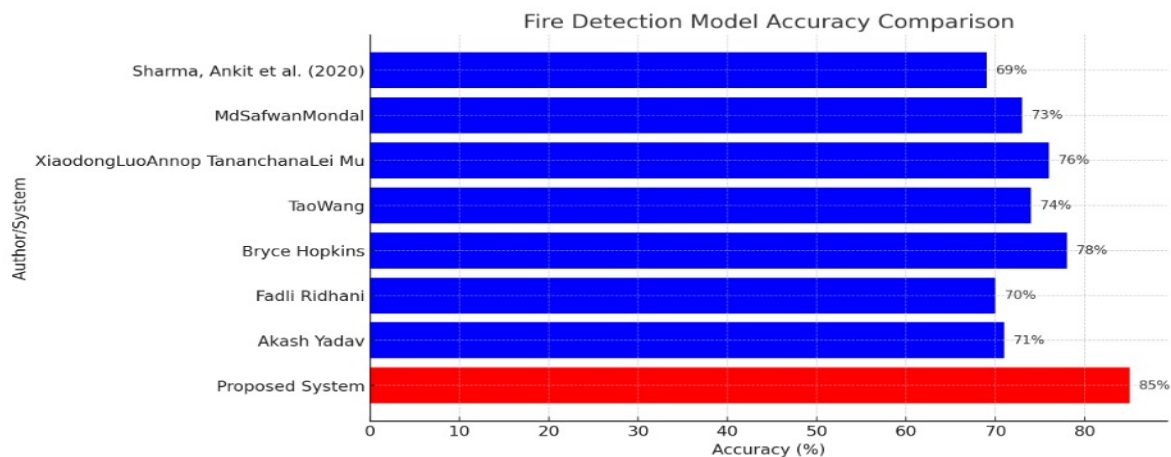


Figure 4.7: Accuracy comparison graph

Vision-Driven Fire Response System utilizing YOLO (You Only Look Once) object detection provides an efficient and accurate approach to real-time fire detection and response. YOLO, known for its ability to process images in a single pass, enables rapid identification of fire in video streams, making it highly suitable for fire surveillance applications. By training the model on a dataset of annotated fire images, it learns to recognize fire characteristics such as shape, texture, and color patterns, ensuring precise localization and detection. The system effectively identifies fire across various conditions, including different lighting environments, fire sizes, and complex backgrounds, ensuring reliable performance in real-world scenarios. Once fire is detected, the system triggers an automated response using a hardware setup that includes an Arduino, an L298 motor driver, a pump, and a servo motor. The Arduino processes the fire detection signal and controls the response mechanism, activating the pump and servo motor to direct water or fire suppressant toward the affected area. This automated action ensures that the fire suppression process starts immediately after detection, minimizing potential damage and preventing further spread. The integration of object detection with a hardware-driven response makes the system more effective than traditional manual fire detection and extinguishing methods. **Figure 4.7** shows a comparison of the proposed system with existing models, highlighting its advantages in terms of model used. Additionally, **Table 4.1** presents a detailed comparison of the proposed model against other fire detection models.

In summary, the Vision-Driven Fire Response System combines YOLO-based fire detection with an Arduino-controlled response mechanism to provide a fast, reliable, and automated solution for fire safety. The system's ability to detect fire in real-time and activate suppression mechanisms enhances safety measures, reducing damage and improving emergency response efficiency. Its adaptability to various environments makes it a valuable tool for residential,

industrial, and public safety applications, ensuring a proactive approach to fire hazard management.
Discussion

CHAPTER 5

CONCLUSION & FUTURE SCOPE

5.1 Conclusion

The Vision-Driven Fire Response System leveraging YOLOv5 represents a significant advancement in fire safety and automated firefighting. By integrating deep learning and real-time vision-based detection, this system ensures rapid identification of fire incidents, minimizing response times and reducing potential damage. The combination of YOLOv5's high detection accuracy with Arduino-controlled fire suppression mechanisms allows for a seamless and automated response, making it an ideal solution for fire-prone environments such as factories, warehouses, and residential spaces.

The system's scalability and adaptability allow it to be deployed across various locations with minimal modifications, ensuring widespread usability. The ability to operate under diverse lighting conditions and environmental challenges further enhances its reliability. Additionally, the integration with a motorized servo system and water pump ensures a precise and targeted suppression of fire, reducing water wastage while improving efficiency.

With continuous advancements in artificial intelligence, computer vision, and embedded systems, future enhancements to YOLO-based fire detection will further improve accuracy, reduce false positives, and enhance automation in firefighting. Overall, this project paves the way for smart, autonomous fire detection systems that can significantly contribute to safety and disaster management efforts. .

5.2 Future Scope

The future scope of the Vision-Driven Fire Response System is vast, driven by advancements in AI-based fire detection, robotics, and IoT-based automation. As research in these fields progresses, several areas of improvement and expansion emerge:

Enhanced Accuracy & Detection Speed Future iterations of YOLO and deep learning models will further refine fire detection capabilities, improving accuracy and reducing false alarms. Enhancements in thermal imaging integration could allow detection even in extreme environmental conditions such as smoke-filled rooms or nighttime scenarios.

Integration with IoT and Smart Buildings The system can be integrated with IoT networks, allowing for automated alerts to be sent to emergency responders, firefighters, and building management systems. This would ensure a more efficient and coordinated response to fire incidents.

Expanded Suppression Techniques Future developments may allow for the incorporation of multiple fire suppression techniques, such as CO₂-based suppression, foam dispensers, or powder-based extinguishers, depending on the type of fire detected.

Wireless and Remote Monitoring Implementing wireless communication protocols (such as Wi-Fi, LoRa, or 5G) would allow real-time monitoring and control from remote locations. Fire incidents could be viewed via a web dashboard or mobile application, improving situational awareness.

Improved Hardware Efficiency Optimizing the Arduino-controlled actuator mechanisms for faster response times and energy-efficient operations would improve the overall performance and longevity of the system. The use of edge computing for real-time inference could further reduce latency and improve processing efficiency.

Advanced AI-Based Fire Suppression Decision Making Integrating machine learning-based decision systems could allow the model to analyze fire intensity and spread direction in real time, dynamically adjusting the water jet's direction, force, and coverage area accordingly.

Scalability for Large-Scale Applications This system can be expanded for industrial and urban applications, including factories, oil refineries, smart cities, and commercial buildings. The integration of drones equipped with fire detection cameras can provide aerial monitoring and automated suppression, making fire response even more efficient and proactive.

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